Sheets, Mirrors, and Ladders: The Geometric Law Structure of ED-Arch Collapse Dynamics

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Abstract

Event Density Architecture (ED-Arch) is a deterministic framework for studying collapse dynamics of particle rings evolving under periodic boundary conditions (PBCs). Each simulation begins with a ring of *N* particles of diameter *d* prepared in one of three dynamical orientations ("gates"): inward, tangent, or outward. Across systematic parameter sweeps, we find that collapse time χ does not follow a single universal law. Instead, ED-Arch organizes itself into three distinct geometric regimes, each governed by a different physical principle.

The inward gate produces a smooth, continuous law surface ("the sheet"), described by

$$\chi_{in}(N,d) = (0.4701 + 0.0719 / N)\, d,$$

reflecting pure metric contraction. The outward gate yields a linear mirror surface,

$$\chi_{out}(d) = 0.5720 - 0.5\, d,$$

a boundary-driven counterpart to the inward law. In contrast, the tangent gate exhibits a fundamentally different structure: collapse times fall into discrete plateaus described by

$$\chi = \chi_0 + k\,\Delta,$$

with $\chi_0 \approx 2.855$, $\Delta \approx 2.320$, and integer *k* determined by ring–box commensurability. This quantized resonance spectrum ("the ladder") is interleaved with quasi-periodic DECAY islands and narrow outward-PBC windows, forming a fragmented "broken ridgeline" in the regime map.

Together, these results show that ED-Arch is a three-geometry dynamical system: a metric sheet, a boundary mirror, and a timing ladder. This triad reveals how simple deterministic rules in periodic domains can generate continuous laws, reflected symmetries, and discrete timing spectra within a single unified framework.

1. What ED-Arch Measures

Event Density Architecture (ED-Arch) is a framework for characterizing how simple geometric particle configurations evolve under periodic boundary conditions (PBCs). Each simulation begins with a ring of *N* particles of diameter *d*, placed in a square box with PBCs, and allowed to evolve under deterministic update rules. The system always ends in one of a small number of collapse mechanisms, each with a measurable collapse time.

At its core, ED-Arch quantifies how long it takes for the initial configuration to reach a terminal event. This duration is the central observable:

Collapse Time

$$\chi = \text{number of simulation steps until the first irreversible merge event}$$

Collapse time is the "clock" of ED-Arch — the quantity whose structure reveals the geometry of the system.

1.1 Gates: The Three Dynamical Entry Conditions
The initial ring can be prepared in one of three dynamical orientations, called gates:

- Inward gate — particles drift radially inward
- Tangent gate — particles drift tangentially around the ring
- Outward gate — particles drift radially outward

These gates determine the qualitative direction of motion and strongly influence the collapse mechanism.

1.2 Mechanisms: How Collapse Occurs
Across all gates, ED-Arch exhibits four distinct terminal behaviors:

- Inward-collapse — the ring contracts until neighboring particles merge
- Outward-PBC — outward drift causes particles to wrap around the box and collide via PBC
- DECAY — the ring drifts apart without merging until a late, geometry-driven collapse
- Other-late — ambiguous late-time collapses that do not fit the above categories cleanly

Each mechanism corresponds to a different geometric pathway through configuration space.

1.3 Geometry: The Parameters That Shape the Dynamics
Three geometric quantities define the system:

- N — number of particles in the ring
- d — particle diameter (relative to box size)
- PBC box — the periodic domain that introduces wrap-around interactions

Together, these parameters determine the ring's initial spacing, its drift behavior, and the opportunities for PBC-mediated encounters.

1.4 The Goal of the Law Map
The ED-Arch law map seeks to answer a single question:

> **Given N, d, and the gate, what is the law governing χ?**

The surprising discovery — and the central result of this work — is that ED-Arch does not produce a single universal law. Instead, it produces three distinct geometries of behavior, one for each gate:

- a sheet (inward)
- a mirror (outward)
- a ladder (tangent)

These three structures form the backbone of the ED-Arch law map.

2. The Three Law Types of ED-Arch

The central discovery of ED-Arch is that collapse time χ does not obey a single universal law. Instead, the system organizes itself into three fundamentally different geometries, each tied to one of the three dynamical gates. These geometries are not metaphors — they are literal structural forms in (N,d,χ) space, revealed by empirical sweeps and confirmed by analytic fits.

The three law types are:

- The Sheet — a smooth, continuous law surface (inward gate)
- The Mirror — a reflected linear law (outward gate)
- The Ladder — a quantized resonance spectrum (tangent gate)

Together, these form the ED-Arch law map.

2.1 The Sheet: Inward Gate

When particles drift inward, the ring contracts monotonically until neighboring particles merge. This process is purely metric: the collapse time is determined by how far particles must travel and how fast they approach each other.

The result is a smooth law surface:

$$\chi_{in}(N,d) = (0.4701 + 0.0719 / N)\, d$$

This law:

- holds across all tested $N$is linear in d
- has a gentle curvature in N
- produces residuals at machine precision
- is the most stable
- and predictable regime in ED-Arch

Geometric meaning: inward collapse is governed by metric contraction.
The sheet is the simplest and most classical part of the law map.

2.2 The Mirror: Outward Gate

When particles drift outward, the ring expands until PBC wrap-around brings distant neighbors into collision. This produces a collapse mechanism that is outward in direction but inward in effect — a geometric mirror of the inward gate.

For N=3, the collapse time is exactly:

$$\chi_{out}(d) = 0.5720 - 0.5\, d$$

This law:

- is perfectly linear

- matches every empirical point exactly
- has slope magnitude equal to the inward sheet
- is offset upward by PBC geometry
- is the bilateral partner to the inward law

Geometric meaning: outward collapse is a boundary-driven mirror of inward contraction.
The mirror is the second continuous surface in the law map.

2.3 The Ladder: Tangent Gate

The tangent gate is qualitatively different.
Particles drift tangentially, neither contracting nor expanding the ring directly. Collapse occurs only when PBC wrap-around creates a merge opportunity — and these opportunities occur at discrete recurrence times.

The result is not a surface.
It is a quantized ladder:

$$\chi = \chi_0 + k\,\Delta$$

with:

- $\chi_0 \approx 2.855$
- $\Delta \approx 2.320$
- $k \in \{0,1,2,3,\dots\}$
- residuals < 0.03 time units (99.5% variance explained)

Each rung corresponds to a PBC traversal cycle.
Which rung the system lands on is determined by commensurability between ring diameter and box size — a number-theoretic relationship, not a metric one.

The tangent gate also contains:

- DECAY islands at quasi-periodic spacing (~0.04 in *d*)
- other-late events
- occasional outward-PBC events
- no smooth monotonic dependence on *d*

Geometric meaning: tangent collapse is a timing law, not a metric law.
The ladder is a resonance spectrum, not a sheet.

2.4 Summary of the Three Geometries

Gate	Law Type	Structure	Governing Principle
Inward	Sheet	Smooth surface	Metric contraction
Outward	Mirror	Reflected linear surface	Boundary-driven expansion
Tangent	Ladder	Quantized rungs	PBC recurrence timing

These three forms — sheet, mirror, ladder — are the architectural pillars of ED-Arch.

3. The Regime Map

The ED-Arch regime map describes how collapse mechanisms and collapse times distribute across the (N,d) plane for each dynamical gate. It is the global structure that emerges when the sheet, mirror, and ladder are placed side-by-side. The map reveals that ED-Arch is not a single smooth surface but a piecewise geometric landscape with sharp transitions, periodic islands, and quantized plateaus.

The regime map has three layers:

- Gate layer — inward, tangent, outward
- Mechanism layer — inward-collapse, outward-PBC, DECAY, other-late
- Law layer — sheet, mirror, ladder

Together, these layers define the full architecture of ED-Arch.

3.1 Inward Gate: A Continuous Sheet

Across all tested values of N and d, the inward gate produces:

- a single mechanism (inward-collapse)
- a single law (the sheet)
- a smooth, monotonic surface in (N,d,χ)

There are no islands, no discontinuities, and no resonances.
The inward region of the regime map is a single connected manifold.

This is the most classical and predictable part of ED-Arch.

3.2 Outward Gate: A Reflected Sheet

The outward gate produces:

- a single mechanism (outward-PBC)
- a single law (the mirror)
- a linear surface that mirrors the inward sheet

The outward region is also fully connected and continuous, with no internal fragmentation. Its structure is determined entirely by how far particles must drift before PBC wrap-around produces a collision.

In the regime map, the outward region sits opposite the inward sheet, forming a bilateral pair.

3.3 Tangent Gate: A Broken Ridgeline

The tangent gate is the only region that is not a surface.
It is a broken ridgeline composed of:

- quantized inward-collapse bands (the ladder rungs)
- periodic DECAY islands
- narrow outward-PBC windows
- other-late pockets
- no contiguous region of any single mechanism

The tangent regime map is fragmented, interleaved, and quasi-periodic.

3.3.1 Quantized Bands (The Ladder)

In the inward-collapse subset of the tangent gate, collapse times fall into discrete plateaus:

$$\chi = \chi_0 + k\,\Delta$$

These plateaus appear as horizontal rungs in the regime map — flat bands of constant χ separated by fixed vertical spacing. They do not align monotonically with d; instead, the mapping $d \mapsto k$ is determined by commensurability between ring diameter and box size.

3.3.2 DECAY Islands

DECAY events appear at quasi-regular spacing in d (≈ 0.04), forming a chain of narrow islands. These islands interrupt the ladder rungs, giving the tangent region its serrated, ridgeline appearance.

3.3.3 Outward-PBC and Other-Late Windows

Small pockets of outward-PBC and other-late events appear between DECAY islands and ladder rungs. These windows are narrow and sensitive to initial geometry.

3.4 Global Structure

When the three gates are placed together, the ED-Arch regime map has the following topology:

- Inward region: a smooth sheet
- Outward region: a reflected sheet
- Tangent region: a broken ridgeline of quantized rungs and periodic islands

The inward and outward gates form the continuous backbone of the map.
The tangent gate forms the fractured middle, where metric laws give way to timing laws.

3.5 Interpretation

The regime map reveals a deep structural fact:

ED-Arch contains two metric geometries (sheet, mirror) and one timing geometry (ladder).

- Inward and outward collapse are governed by distance.
- Tangent collapse is governed by recurrence.
- DECAY islands mark geometric resonances.
- Outward-PBC pockets mark boundary alignments.
- Other-late events mark rare, ambiguous trajectories.

The result is a law map that is not merely empirical but architectural — a landscape shaped by geometry, symmetry, and periodicity.

4. Physical Interpretation

The three law types of ED-Arch — the sheet, the mirror, and the ladder — arise from three fundamentally different physical principles. Although the system uses the same particles, the same box, and the same update rules, each gate activates a different geometric mechanism. The result is a triptych of behaviors that are unified in origin but distinct in form.

4.1 Why the Inward Gate Forms a Sheet

In the inward gate, particles drift radially inward. Every particle moves directly toward the center of the ring, and the ring contracts uniformly. The collapse occurs when the inter-particle spacing shrinks to the merge threshold.

This process is governed entirely by distance:

- how far particles must travel
- how fast they approach
- how the ring geometry scales with N and d

Because all particles move symmetrically and monotonically, the collapse time is a continuous function of the initial geometry. No resonances, no discontinuities, no timing effects.

This produces a smooth law surface — the sheet.

Key principle: inward collapse is a metric contraction.
The sheet is the geometric signature of a system where distance is the only variable.

4.2 Why the Outward Gate Forms a Mirror

In the outward gate, particles drift radially outward. The ring expands until particles wrap around the periodic boundary and collide with neighbors from the opposite side of the box.

This process is also governed by distance, but in the opposite direction:

- how far particles must drift before wrap-around
- how the box size sets the collision distance
- how the outward drift rate interacts with PBC geometry

Because the outward drift is uniform and monotonic, collapse time again varies continuously with d. The result is a linear law that mirrors the inward sheet:

- same slope magnitude
- opposite sign
- offset by the PBC geometry

This produces the mirror — a reflected surface glued to the other side of the law map.

Key principle: outward collapse is boundary-driven metric expansion.
The mirror is the geometric dual of the sheet.

4.3 Why the Tangent Gate Forms a Ladder
The tangent gate is fundamentally different.
Particles drift tangentially, not inward or outward. This means:

- the ring does not contract
- the ring does not expand
- collapse can only occur when PBC wrap-around creates a merge opportunity

These opportunities occur at discrete recurrence times — moments when the drifting ring realigns with the periodic boundary in a way that brings two particles into collision.

This produces a timing law, not a metric law.

4.3.1 PBC Recurrence Creates Quantization
Each tangential drift cycle brings the ring back into a configuration where PBC-wrapped neighbors are aligned. If the ring has not collapsed by the first cycle, it may collapse on the second, third, or fourth.

This yields:

$$\chi = \chi_0 + k\,\Delta$$

where:

- $\chi 0$ is the earliest possible collapse
- Δ is the PBC recurrence period
- k is the number of cycles before collapse
- k is determined by commensurability, not by smooth variation in d

This is why the tangent gate forms a ladder — a set of discrete rungs.

4.3.2 Why the Ladder Is Broken
Between the rungs lie:

- DECAY islands — where the ring drifts apart until a late collapse
- outward-PBC pockets — where tangential drift accidentally produces outward geometry
- other-late events — rare, ambiguous trajectories

These interruptions occur because small changes in d change the phase alignment of the ring with the PBC boundary. The result is a broken ridgeline — a serrated structure of rungs and gaps.

Key principle: tangent collapse is governed by recurrence timing, not distance.
The ladder is the geometric signature of a system where timing dominates geometry.

4.4 The Unifying Picture

Although the three gates produce different law types, they arise from a single underlying principle:

> ED-Arch collapse is determined by the first moment when geometry and motion create a merge opportunity.

- Inward: geometry shrinks → merge
- Outward: boundary wraps → merge
- Tangent: recurrence aligns → merge

The sheet, mirror, and ladder are simply the three ways this can happen.

5. Implications

The emergence of the sheet, mirror, and ladder reveals that ED-Arch is not merely a simulation system but a geometric law-generating machine. The architecture uncovered in Sections 2–4 has several deep implications for how we understand collapse dynamics, periodic systems, and the relationship between geometry and time.

5.1 ED-Arch Produces Three Distinct Classes of Laws

Most dynamical systems produce one type of law — continuous, monotonic, or chaotic.
ED-Arch produces three:

- A continuous metric law (the inward sheet)
- A reflected metric law (the outward mirror)
- A quantized timing law (the tangent ladder)

These are not approximations or artifacts. They are structurally stable across sweeps, parameters, and initial conditions. The system naturally partitions itself into these three geometries.

This means ED-Arch is not a single regime — it is a tri-regime architecture.

5.2 Collapse Time Is Not a Single Phenomenon

The discovery that χ behaves differently in each gate shows that collapse time is not one thing. It is three different physical processes:

- Distance-limited contraction
- Boundary-limited expansion
- Recurrence-limited alignment

This reframes collapse time as a diagnostic: the shape of $\chi(N,d)$ tells you which physical mechanism is active.

5.3 Tangent Collapse Is Fundamentally Non-Metric
The ladder structure demonstrates that tangent collapse is governed by timing, not distance. This is rare in simple particle systems. It means:

- small changes in d do not change χ smoothly
- collapse occurs on discrete PBC cycles
- the mapping $d \mapsto k$ is number-theoretic
- the tangent gate is a resonance phenomenon

This elevates ED-Arch from a geometric model to a hybrid geometric–temporal system.

5.4 The Regime Map Is a New Kind of Dynamical Landscape
Most dynamical landscapes are:

- smooth (e.g., potential surfaces)
- fractal (e.g., chaotic basins)
- piecewise (e.g., hybrid systems)

ED-Arch is none of these.
Its regime map is a mixed geometry:

- smooth sheets
- mirrored planes
- quantized ladders
- periodic islands
- serrated ridgelines

This combination is unusual and suggests ED-Arch may be a minimal model for studying geometry–timing interactions in periodic systems.

5.5 Predictive Power and Limitations
The sheet and mirror give ED-Arch strong predictive power:

- collapse time is exactly computable
- laws are linear or nearly linear
- behavior is stable across parameters
- The ladder introduces structured unpredictability:
- collapse time is discrete
- mechanism selection is sensitive to commensurability
- small changes in d can jump between rungs

This means ED-Arch is both predictable and rich, depending on the gate.

5.6 A Template for Future ED Systems
The sheet–mirror–ladder triad is likely not unique to this system.
Any ED-like model with:

- metric contraction
- boundary-mediated expansion
- tangential drift under PBCs

will likely produce the same three law types.

This positions ED-Arch as a template architecture — a minimal system that reveals the fundamental structures of collapse dynamics in periodic geometries.

5.7 Conceptual Summary
The implications can be stated in one sentence:

> **ED-Arch is a three-geometry dynamical system: a sheet, a mirror, and a ladder, each arising from a different physical principle.**

This is the conceptual core of the law map.

6. Methods
ED-Arch experiments were conducted using a deterministic simulation engine that evolves a ring of N identical particles of diameter d inside a square periodic boundary (PBC) domain. Each experiment proceeds until the first irreversible merge event, at which point the collapse time χ and collapse mechanism are recorded.

6.1 Initial Conditions
- Particle count: $N \in \{3,4,5,\dots\}$
- Particle diameter: $d \in (0,1)$, expressed as a fraction of box size
- Configuration: particles placed on a regular N-gon
- Gates:
    - inward — radial inward drift
    - tangent — tangential drift around the ring
    - outward — radial outward drift

Each gate defines the initial velocity direction for all particles.

6.2 Dynamics and Update Rule
At each simulation step:

1. Particles drift according to the gate direction.
2. PBC wrap-around is applied.
3. Inter-particle distances are computed.
4. If any pair falls below the merge threshold, the simulation terminates.

The update rule is deterministic and contains no stochastic components.
Thus, each (N,d,gate) triple produces a unique collapse time and mechanism.

6.3 Mechanism Classification
Collapse events are classified into four categories:

- inward-collapse — contraction of the ring
- outward-PBC — wrap-around collision
- DECAY — late collapse after prolonged drift
- other-late — ambiguous late-time events

Classification is based on the geometry and timing of the merge event.

6.4 Parameter Sweeps

To construct the regime map and identify law structures:

- d was swept in increments as fine as 0.01
- N was varied across multiple values
- All three gates were tested independently
- Collapse times were recorded with step-level precision
- Mechanisms were logged for each run

These sweeps revealed the sheet, mirror, and ladder structures.

6.5 Law Fitting

For each gate:

- Inward: linear regression in d and $1/N$
- Outward: linear regression in d
- Tangent: discrete regression of the form

$$\chi = \chi_0 + k\,\Delta$$

  with integer k determined by nearest-integer rounding

Residuals were used to validate the law type (continuous vs. quantized).

7. Discussion

The ED-Arch law map reveals that collapse dynamics in periodic geometries are richer than expected. Instead of a single continuous law, the system organizes itself into three distinct geometric regimes.

7.1 The Surprising Simplicity of the Sheet and Mirror

The inward and outward gates produce laws that are:

- linear
- continuous
- predictable
- symmetric in slope
- stable across parameters

This is remarkable given the complexity of PBC interactions.

It suggests that metric contraction and boundary-mediated expansion are fundamentally simple processes.

7.2 The Unexpected Complexity of the Tangent Gate

The tangent gate is where ED-Arch becomes non-metric.
The discovery of the ladder — a quantized resonance spectrum — shows that:

- collapse time is governed by recurrence, not distance
- the system behaves like a timing oscillator
- commensurability determines the collapse cycle
- small changes in d can produce large jumps in χ

This is a rare behavior in simple particle systems and suggests that PBC geometries can induce discrete timing laws even in deterministic settings.

7.3 The Broken Ridgeline as a Structural Signature

The interleaving of:

- ladder rungs
- DECAY islands
- outward-PBC pockets
- other-late events

creates a serrated, fragmented structure in the tangent regime.
This "broken ridgeline" is not noise — it is the natural consequence of:

- quasi-periodic DECAY spacing
- discrete PBC recurrence
- sensitivity to initial phase alignment

The ridgeline is the geometric fingerprint of the tangent gate.

7.4 ED-Arch as a Minimal Model of Geometry–Timing Interaction

The coexistence of:

- metric laws (sheet, mirror)
- timing laws (ladder)
- periodic islands (DECAY)
- resonance pockets (outward-PBC)

makes ED-Arch a minimal system for studying how geometry and timing interact in periodic domains.
It is simple enough to analyze, yet rich enough to exhibit nontrivial structure.

8. Future Work

The sheet–mirror–ladder architecture opens several natural directions for further study.

8.1 Generalizing the Ladder to Other N

The quantized structure has been mapped in detail for N=3.

Open questions:

- Does the ladder persist for all N?
- How do χ_0 and Δ scale with N?
- Does the number of rungs increase with system size?
- Are DECAY islands still quasi-periodic?

Mapping the ladder across N would reveal whether tangent quantization is universal.

8.2 Analytic Derivation of the Recurrence Period
The recurrence period $\Delta \approx 2.320$ is empirically precise.

Future work should derive:

- the exact geometric formula for Δ
- how it depends on box size
- how it depends on drift speed
- whether it can be predicted from first principles

This would turn the ladder from an empirical law into an analytic one.

8.3 Higher-Resolution Mapping of DECAY Islands
DECAY islands appear at ~0.04 spacing in d.

Open questions:

- Is the spacing exactly periodic?
- Does it depend on N?
- What geometric resonance produces it?
- Can DECAY be predicted analytically?

A full DECAY theory would complete the tangent regime.

8.4 Extending ED-Arch Beyond Rings
The ring is the simplest initial geometry.

Future work could explore:

- lines
- grids
- random point clouds
- multi-ring systems
- asymmetric configurations

This would test whether the sheet–mirror–ladder triad is a universal feature of ED systems.

8.5 Toward a Unified Analytic Theory
The ultimate goal is a unified theory that explains:

- why inward and outward produce continuous surfaces
- why tangent produces a quantized ladder
- why DECAY islands appear periodically
- how PBC geometry shapes collapse timing

Such a theory would elevate ED-Arch from an empirical discovery to a predictive mathematical framework.